



6-Month Full AI/ML + Django Roadmap



MONTH 1 — Python + Data Foundations

Learn

- Python (loops, functions, OOP)
- Numpy
- Pandas
- Matplotlib / Seaborn
- Git + GitHub
- Basic Linux commands
- Jupyter Notebook

Understand

- Data cleaning
- Feature engineering
- Basic EDA (graphs, correlations)

Projects

- Netflix EDA
- Sales prediction data analysis
- Titanic survival analysis
- 10 Python mini-automation scripts

Output: Strong Python + Data handling



MONTH 2 — SQL + Machine Learning

SQL (1 week)

- SELECT, WHERE
- GROUP BY
- JOINS
- Subqueries
- Indexes
- PostgreSQL/MySQL basics

Machine Learning (3 weeks)

- Linear Regression
- Logistic Regression

- Decision Trees
- Random Forest
- KNN
- SVM
- Naive Bayes
- K-means
- PCA
- Train/validation/test methods
- Hyperparameter tuning

Projects

- Loan default prediction
- House price prediction
- Customer segmentation
- Spam detector
- Credit card fraud detection

Output: 5 ML projects + solid SQL

MONTH 3 — Django + Backend Development

Learn Django

- Models
- Views
- Templates
- URL routing
- Forms
- User authentication
- Django Admin
- Django REST Framework (DRF)

Build

- Login + Signup system
- Blog app
- REST API using DRF
- File upload API

Output: Backend + API development skills

MONTH 4 — ML + Django Integration + Deployment Basics

Integrate ML with Django

- Save ML model (.pkl)
- Load model inside Django
- Prediction API
- Web form → prediction output
- Secure endpoints

Deployment Basics

- Basic Linux
- Nginx
- Gunicorn
- PostgreSQL (production)
- Environment variables

Projects

- ML prediction web app (price prediction/medical prediction)
- Admin dashboard
- Public API endpoints

Output: Real AI website + deployment knowledge

MONTH 5 — Deep Learning + NLP + LLM Basics

Deep Learning

- Neural Networks
- PyTorch
- CNN
- Transfer Learning (ResNet/MobileNet)
- RNN/LSTM

NLP

- Tokenization
- Word embeddings (Word2Vec, GloVe)
- Transformers (BERT)
- Text classification

LLM Basics

- Embeddings
- Prompt engineering

- RAG basics
- HuggingFace

Projects

- Cats vs Dogs classifier
- Emotion detection
- News classifier with BERT
- Simple chatbot

Output: Strong DL + NLP + Transformer basics



MONTH 6 — RAG, Vector DB, MLOps + Final Portfolio

Learn

- Vector Databases (FAISS/Pinecone)
- RAG architecture
- LangChain
- FastAPI for ML/AI
- Docker
- CI/CD basics
- Cloud deployment (Render/Railway/AWS)

Final Portfolio Projects

1. AI + Django SaaS App

Examples: - PDF chatbot - AI content generator - Resume analyzer

2. Deep Learning Product

- Image classifier web app
- Emotion recognition API

3. RAG App with Vector DB

- PDF Q&A system
- Multi-document search AI

Final Prep

- Resume
- LinkedIn
- GitHub cleanup
- Mock interviews

Output: Job-ready portfolio with real AI products



Skills Achieved After 6 Months

- Python
- SQL
- Django + DRF
- Machine Learning
- Deep Learning
- NLP + Transformers
- LLMs + RAG
- Vector Databases
- Deployment (Docker, FastAPI, Cloud)
- Full AI product development